PUNJAB
FLOODPLAIN ENCROACHMENT REMOVAL

Sector: Infrastructure & Urban Resilience
Hazard adressed: Flood
Targeted assets: Roads, Bridges, Main Canal, Minor Canal, Distributaries, Communication Towers, Residential Buildings, Industrial Buildings, Commercial Buildings, Education Facilities, Health Facilities

Description: This measure involves the removal of building encroachments from within natural waterways, restoring floodplains to their original widths and improving flood conveyance capacity. Encroaching residential buildings are identified within a 500m waterway corridor; owners are compensated at 30% of building value (building only, land excluded). Afforestation and wetland restoration components are excluded from scope. Post-removal maintenance is through law enforcement only, with no OPEX.

Priority Districts: Mandi Bahauddin, Gujrat, Narowal, Sheikhupura, Nankana Sahib, Lahore

TECHNICAL SPECIFICATIONS/DIMENSION

Unit of implementation: Per sq. km of floodplain restored.

Scale assumptions: 2% of total floodplain area in the target districts considered; 500m waterway width assumption.

Key design parameters: Waterway corridor width = 500m; compensation = 30% of building value (building only, land excluded); only residential buildings included; post-removal maintenance via law enforcement.

Time horizon: 2026–2030.

IMPACT

Risk reduction level: Very High Effectiveness – large reduction in damages (Expert Rank 1).

Time to effectiveness: Effectiveness is progressive as encroachments are removed.

Expected impact: 20% flood intensity reduction; applicable for 1-in-100-year flood events; MDD reduced to 80% of original across all asset classes within restored floodplains.

IMPLEMENTATION CONSIDERATIONS

Enabling conditions: Legal authority for encroachment removal; compensation fund availability; building asset surveys and valuation; political will; community engagement and grievance redressal mechanisms.

Required institutions: Punjab Irrigation Department; Revenue Department (for asset valuation and compensation); Punjab Land Record Authority; local law enforcement agencies for initial removal and ongoing maintenance.

Barriers: Political resistance; legal challenges; insufficient compensation funding; risk of re-encroachment without sustained enforcement; limited scale (only 2% of floodplains) limits overall impact.

COST ASSUMPTIONS

Unit cost: USD 3,454,917 per sq. km.

Capital Expenditure (CAPEX): USD 19,323,482.

Operational Expenditure (OPEX): No OPEX (=0) post-removal maintenance through law enforcement only).

CO-BENEFITS & TRADE-OFFS

Environmental: Restoration of natural hydrological connectivity; riparian habitat recovery; reduced riverbed scouring; potential wetland and biodiversity regeneration once buildings removed.

Social: Reduced flood risk for downstream communities; reduced risk to lives from flash floods; however potential social disruption from displacement of encroaching residents (managed through compensation).

Economic: Avoided costs of repeated flood damage to infrastructure; reduced burden on disaster relief and emergency response; long-term savings from restored natural flood attenuation.

Potential risks: Significant social resistance from displaced encroachers; legal challenges to removal orders; inadequate compensation leading to litigation; political sensitivity; Only 2% restoration – much larger investment needed for meaningful scale.

BENEFICIARIES

Primary beneficiaries: Downstream communities protected from floods.

Distributional aspects: Net beneficiaries are downstream flood-prone communities; distributional impacts on displaced residents require careful management.

Note: This factsheet is to be used as a rapid planning and appraisal tool. All costs and impacts should be validated through field assessments, stakeholder consultation and detailed feasibility studies.